Robert Burdsall

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Professional Summary

Electrical and Computer Engineering student at NC State with hands-on industry experience in PLC programming, electrical panel design, and machine vision integration for manufacturing automation. Proven ability to lead controls projects from design through deployment, with additional depth in embedded systems, real-time control logic, and software development. Seeking a controls or automation engineering internship in robotics or autonomous systems.

Education

North Carolina State University — Raleigh, NC
Bachelor of Engineering in Electrical and Computer Engineering. GPA: 3.5 — *Aug. 2025 – May 2028*

North Carolina School of Science and Mathematics — Morganton, NC
High School Diploma, Computer Science Focus. GPA: 4.8 — *Aug. 2023 – May 2025*

Experience

Electrical Engineering Intern — May 2026 – Aug. 2026
IDEAL Fastener Corporation — *Oxford, NC*

- Designed and fabricated 5+ electrical control panels for production machinery, handling full wiring assembly including relays, power supplies, DIN rail, disconnects, proximity and laser sensors, and solenoid valves, and produced TinyCAD schematics for future replication.
- Programmed Omron PLCs using CX-Programmer to implement automated machine control logic; configured and trained Keyence IV3/IV4 vision systems by defining inspection criteria and teaching defect recognition to upgrade 3 existing floor machines for real-time quality detection, establishing automated defect detection for the first time on each machine and achieving 100% inspection coverage.
- Led end-to-end design of a zipper cap inspection machine inspecting 120 caps/min, featuring Keyence vision inspection, vacuum ejection, solenoid valve sequencing, a DOPSoft HMI, and PLC-driven quality tracking with alarm functionality for good/bad rate monitoring.

Lead Mentor and Software Developer — Jan. 2024 – Present
FIRST Robotics Competition, NC School of Science and Mathematics, Team 9150 — *Morganton, NC*

- Designed and implemented full robot control software in Java using WPILib, earning 3rd-highest scoring performance across all North Carolina competitions including the 2025 Asheville Regional, with integrated vision and dual scoring systems.
- Mentor 25+ students as Lead Mentor, teaching Java, WPILib, and robotics programming fundamentals.

Computer Science Teaching Assistant — Aug. 2024 – May 2025
North Carolina School of Science and Mathematics — *Morganton, NC*

- Taught 150+ students programming fundamentals in Java and Python.
- Built a cross-platform resource tracking application using Flutter and Dart in collaboration with faculty.

Projects

Aurum | *Dart, Python, Flutter, ExpressJS, TensorFlow* — Sep. 2024 – Dec. 2024

- Developed a cross-platform Flutter mobile application for real-time coin appraisal, leveraging TensorFlow Lite for on-device inference.
- Engineered live coin value retrieval and portfolio tracking features, integrating client-side and server-side logic for seamless data synchronization.
- Implemented backend infrastructure using Express.js, JSON APIs, and JWT-based authentication.

Activities & Leadership

Boy Scouts of America — 2015 – Present
Eagle Scout, Senior Patrol Leader and Assistant Scoutmaster, Troop 222 — *Huntersville, NC*

- Mentor to 30+ Scouts as Assistant Scoutmaster and lead outdoor programs focused on leadership, teamwork, and problem-solving.

Skills

Industrial and Design: CX-Programmer, Keyence SmartNavigator, TinyCAD, DOPSoft, SolidWorks, Prusa Slicer, Cura
Languages: Java, Python, C/C++, Dart, SQL, JavaScript, HTML/CSS
Developer Tools: Git, Docker, VS Code, IntelliJ, TensorFlow, Google Cloud Platform, Android Studio, XCode